

NutriBot

AI-Powered Personalized Nutrition Assistant

IBM watsonx.ai Powered Chatbot

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Problem Statement & Objectives

Why NutriBot?

Bridging the gap between people and reliable, personalized nutritional guidance using the power of Generative AI.

1

The Problem

People lack access to personalized nutrition guidance; online information is often confusing, unreliable, or requires expensive expert consultation.

2

Growing Need

Healthy meal planning demands expert knowledge, and there is a rapidly growing demand for AI-powered intelligent health assistants.

3

Our Objectives

Develop an intelligent nutrition chatbot that delivers personalized dietary recommendations and promotes healthy eating through conversational AI.

System Architecture

Frontend

HTML, CSS & JavaScript user interface for entering queries and displaying AI responses.

Flask Backend

Python-powered Flask server processes requests and bridges the frontend with IBM watsonx.ai API.

IBM watsonx.ai

Granite Foundation Model authenticates via IBM Cloud API and generates intelligent, personalized nutrition responses.

User Interface

HTML, CSS & JavaScript

Flask Backend

Python server and routing

IBM watsonx.ai

API for model inference

Granite AI Response

Processed reply to UI

Technology Stack

Category	Technologies
Frontend	HTML, CSS, JavaScript
Backend	Python, Flask
AI Platform	IBM watsonx.ai, Granite Foundation Model
Configuration	Python Dotenv, REST API
Development	Git, Virtual Environment



Web Frontend

Responsive chat interface for seamless user interaction.



AI Backend

Scalable IBM Cloud-powered intelligence engine.

WORKFLOW

How NutriBot Works

User Query

Frontend Request

Flask Processing

Cloud Auth

NutriBot seamlessly connects the user's question to a powerful AI engine and returns intelligent, personalized nutrition guidance in real time.

Key Capabilities



Natural Language Understanding

Interprets free-text nutrition questions contextually.



Personalized Recommendations

Tailors dietary advice to the individual user.



Real-time AI Responses

Instant answers powered by Granite Foundation Model.



NutriBot — Your Personalized AI Nutrition Coach

 Chat

 Dashboard

 Meal Plan

 BMI

 Family

 Profile



● Mock Mode



NutriBot

Your Personalized AI Nutrition Coach

Powered by IBM Watsonx.ai · Granite Foundation Models

 Start Chatting

 Calculate BMI

 Dashboard

✓ Personalized Plans

✓ Indian Cuisine

✓ Family Profiles

✓ Calorie Analysis

✓ BMI & Macros

✓ Dark Mode



Meal Planner

Generate AI-powered personalized meal plans

Plan Settings

Plan Duration

1 Week

Diet Type

Balanced (All foods)

Primary Goal

Maintain Weight

Regional Cuisine Preference

Pan-Indian (Mixed)

Additional Preferences

e.g., no onion-garlic, quick meals under 30 min, budget friendly...

Your Meal Plan



Configure your preferences and click
"Generate Meal Plan" to get started.



NutriBot

Ask anything about nutrition, health, fitness



Hello! I'm NutriBot, your AI-powered nutrition coach!

I can help you with:

- Personalized meal plans & Indian recipes
- Calorie & nutrition analysis
- BMI, macros & water intake
- Family diet recommendations
- Weight loss / gain strategies
- Supplements & vitamins guidance

What would you like to know or explore today?

Now

Healthy Breakfast

Weight Loss Diet

Veggie Meal Plan

Protein Sources

Immunity Boost

Hydration Tips



Quick Nutrition Tips

- Eat a rainbow — varied colors = varied nutrients
- Dal + rice = complete protein combination
- Drink water before meals to manage portions
- Include seasonal fruits and vegetables daily
- Haldi (turmeric) milk supports immunity

Refresh Tips



Analyze a Food

e.g., 2 chapati with dal and sabzi...

Analyze Nutrition



Your Profile

Edit

Set up profile

Features & Applications

What NutriBot Offers

A versatile AI-powered nutrition platform serving individuals, healthcare professionals, and organizations alike.

Key Features

1

AI Nutrition Assistant

Personalized meal suggestions and healthy diet recommendations via interactive chatbot.

2

Intelligent Responses

Fast, accurate, AI-generated answers with secure IBM Cloud integration.

3

User-Friendly Interface

Simple conversational UI accessible to all users without technical expertise.

Applications

- Personal health management
- Fitness and wellness coaching
- Hospitals and clinics
- Dietitian support tools
- Educational institutions
- Corporate wellness programs

Advantages & Future Enhancements



Current Advantages

Available 24×7 with a user-friendly interface; delivers accurate AI-generated responses on a scalable, cloud-integrated architecture that improves nutrition awareness.



Near-Term Enhancements

Voice-enabled chatbot, multilingual support, image-based food recognition, and a calorie tracking dashboard.



Long-Term Vision

Mobile application, wearable device integration, and personalized health analytics for comprehensive wellness management.

Conclusion & Thank You

Transforming Nutrition Guidance with Generative AI

NutriBot demonstrates how IBM watsonx.ai and the Granite Foundation Model can power an intelligent healthcare chatbot that delivers personalized nutritional guidance through natural language conversations.

Generative AI in healthcare makes nutrition advice more accessible, interactive, and efficient for everyone.

Project Impact

Practical application of Generative AI in healthcare, bridging the gap between users and expert nutritional knowledge.

Technology Showcase

End-to-end integration of IBM watsonx.ai, Granite Foundation Model, Flask, and a responsive web frontend.

Thank You

Questions & Discussion — Presented by Afreen Thasneem, B.E CSE, J.N.N Institute of Engineering.